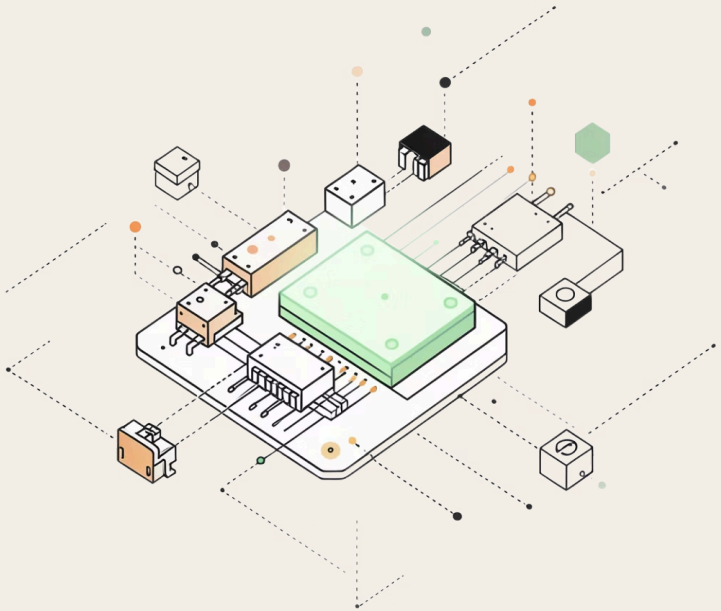


Half Adder & Full Adder Design in Logisim

A hands-on guide to designing and simulating binary adder circuits for students learning digital logic.

Made with GAMMA

Logic Gates: The Building Blocks

XOR Gate

Output is **1** when inputs differ

$A \oplus B$

AND Gate

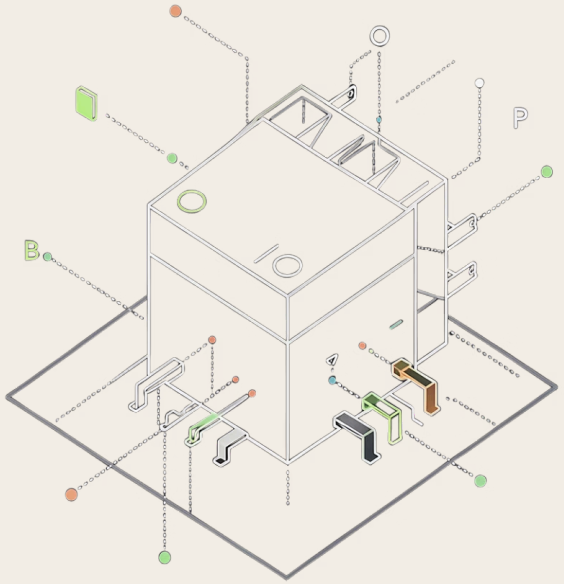
Output is **1** only when both inputs are 1

$A \cdot B$

OR Gate

Output is **1** when at least one input is 1

$A + B$



Made with GAMMA

Half Adder: Adding Two Bits

The half adder is the simplest binary addition circuit, taking two single-bit inputs and producing two outputs.

- **Inputs:** Two single-bit numbers (A, B)
- **Outputs:** Sum (S) and Carry (C)
- **Sum:** A XOR B | **Carry:** A AND B

A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Made with GAMMA

Full Adder: Adding Three Bits

A	B	Cin	Sum (S)	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The full adder extends the half adder by accepting a carry-in from a previous stage — essential for multi-bit arithmetic.

- **Inputs:** Two bits (A, B) plus carry-in (Cin)
- **Outputs:** Sum (S) and Carry-out (Cout)
- **Design:** Two half adders + one OR gate to combine the carries

All 8 possible input combinations are covered — verify each row against your Logisim simulation.

Made with GAMMA

Building in Logisim: Step-by-Step

01

Create Input Pins

Label them **A**, **B**, and **Cin** (for full adder)

02

Add Logic Gates

Drag **XOR**, **AND**, and **OR** gates from the Gates library

03

Wire the Circuit

Connect inputs to gates using wires following the logic equations

04

Create Output Pins

Label outputs **Sum** and **Cout**

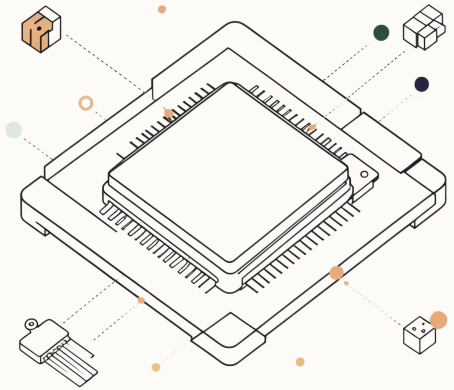
05

Test with Poke Tool

Toggle input values and verify outputs match the truth table

Made with GAMMA

Testing & Applications



Verification

Use Logisim's **Poke tool** to systematically test every row of the truth table. All outputs must match before moving on.

Cascading Adders

Chain full adders together — connecting each **Cout** to the next **Cin** — to build **4-bit** or **8-bit** ripple-carry adders.

Real-World Relevance

Adders are core components of every CPU's **Arithmetic Logic Unit (ALU)**, performing billions of additions per second.

Made with GAMMA

Sources

- **Adder (electronics) — Wikipedia**

Made with GAMMA